

Nihir Praful Shah

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Full Stack Developer with PG-DAC and B.E in CS Degree and total experience of 7+ years with strong grasp in React JS, Next JS, JavaScript, TypeScript, Redux, Node JS, Strapi, MySQL, AWS and Agile Methodology.

Experience

Senior Software Engineer – Kotak Securities, Mumbai OCT 2024 – Present (21 Months)

➤ KEY IMPACT

- Delivered a **major platform modernization** in ~3 months (team of 5) by migrating **Next.js 13 → Next.js 15 (App Router)**, **SSG → ISR**, **JavaScript → TypeScript**, **SCSS + Material UI → Tailwind + shadcn/ui**, and **Strapi v4 → v5**, improving performance, accessibility, scalability and reducing content update time through Strapi CMS.
- Reduced frontend **bundle size by several MBs** by redesigning icons usage with lazy-loaded icon registry, **eliminating ~86% of unused icons**, removing unused packages, lazy loading critical chart libraries.
- Optimized Core Web Vitals for Homepage, Stock & Indices pages serving 120K+ monthly visitors, **reducing LCP** by approximately **32% (4.2s → 2.9s)**, **improving INP** & eliminating **CLS**, resulting in **~8% lower bounce rate**.
- Led **WCAG accessibility** implementation across kotakneo.com per SEBI guidelines, categorizing **1,679+ violations** by severity, coordinating team efforts, and **resolving ~92% of P0/P1 issues** within **3 weeks** to achieve compliance.

➤ TECHNICAL CONTRIBUTIONS

- Architected kotakneo.com using **Next.js & Strapi CMS** with **CloudFront CDN** caching and **Nginx** for load balancing.
- Re-architected Stocks data pipeline by implementing delta-based processing (additions/deletions/updates only), reducing runtime from **16-20hrs to 30-40 mins**. Also built a **Redis caching** layer with **smart pre-caching of ~250 Large/Mid Cap stocks** as they constitute many of the high-traffic pages.
- Optimized application performance by parallelizing ~20 server-side API calls using **Promise.allSettled**, reducing component complexity and unnecessary re-renders, and leveraging **useTransition/startTransition** for non-blocking UI updates, resulting in faster page responsiveness and improved user experience.
- Developed modular, responsive React.js components using custom hooks and **React Server Components** with **Suspense**, integrating **GraphQL** queries for efficient data fetching and optimized rendering performance.
- Designed and developed a **Redis-backed search API** with exact, **tokenized**, and **fuzzy** matching capabilities, supporting **Stocks, Indices, ETFs, and Mutual Funds**. Implemented custom relevance **scoring and ranking** logic to deliver fast, accurate search results across global search and listing pages.
- Implemented **GraphQL retry with exponential backoff and SSR fallback**, improving resilience during API failures.
- Applied **Nginx-based crawler routing**, diverting bots to dedicated server to protect prod performance during spikes.
- Architected a **centralized Redis-based** caching solution (primary + read-replica) with a custom **Next.js cache-handler** to resolve cache inconsistency across multi-server deployments.
- Designed a **webhook-driven cache invalidation** system using Strapi and **unstable_cache**, eliminating the need for full redeployments across multiple servers whenever shared header/footer content changes.
- Implemented **custom Strapi plugin** to translate blogs into **Hinglish**, preserving domain-specific terminology.

Senior Software Engineer - DemandHelm, Mumbai MAR 2021 – SEPT 2024 (44 Months)

➤ Senior Software Engineer (Jan 2023 – Sep 2024)

- Revamped **Sign-Up/Login process** using **AWS Cognito**, introducing **TOTP-based MFA** for enhanced security.
- Migrated deployment from **EC2 + PM2** to a scalable **S3 + CloudFront** setup, enhancing **CI/CD Pipelines** with **Bitbucket Pipelines** with AWS integration for seamless deployments.
- Developed frontend flows for **Amazon Advertising Campaign Creation** using **custom hooks**, **React Hook Form**, and **Zod** for schema validation and comprehensive error handling.
- Led backend development for **Walmart Ads Sign-up** process via **OAuth**, integrating with 3rd-party APIs via **CRON jobs**, **AWS SQS**, and a **Data Processor** for automated daily/weekly sync.
- Engineered robust REST APIs enabling direct management of **Amazon & Walmart Campaigns, Keywords, Ad-Groups, and Ad-Items**, with **AJV schema validation** and **TypeORM** for reliable database operations.

- Added **CloudWatch logging & monitoring** with **OpenTelemetry**. Set up **Slack alerts** for proactive failure detection.
- Implemented unit and integration tests using **Vitest (Jest-compatible API)** with **Storybook** and **Chromatic** for visual regression testing across shared component libraries.

➤ **Software Engineer (March 2021 – Dec 2022)**

- Developed reusable functional components using **Styled components**, **TypeScript & React hooks** and creating webpages from **Figma** designs. Managing the global state and API calls using **Redux Toolkit Query (RTK Query)**.
- Created stunning data visualizations in using **Chart JS & ReCharts** and **Tanstack table**. Added **Virtualization** in tables to enhance performance by rendering only visible rows and avoiding over-allocation of DOM nodes.
- Developed a mobile App as **POC using React Native** that supported login feature. Integrated **Expo Push Notification** service to trigger notifications to registered users on app whenever data was dumped into **AWS S3**.

Systems Engineer – Infosys JUNE 2018 – FEB 2020 (20 Months)

➤ **Systems Engineer (June 2018 – Feb 2020)**

Pune, India

- Developed & architected a **Python automation Sikuli** script to automate daily DBA user management tasks.
- Improved resolution SLA from **98.15% to 98.85%** & reduced manual workload by **120 hours per month**.

Education

(FEB 2020 – FEB 2021) – **Official percentage: 83.9%**

Post-Graduation Diploma in Advanced Computing (PG-DAC)

Vidyanidhi Infotech, Mumbai

COMPUTER SCIENCE (2014- 2018) - **CGPA: 7.91 / 10**

Bachelor of Engineering

Thakur College of Engineering and Technology (TCET), Mumbai University

Skills

- | | | |
|-----------------------------------|---------------------------|--------------------------|
| • React JS | • Next JS | • Lambda, S3 |
| • JavaScript (ES6) & TypeScript | • GraphQL, REST API | • EC2, Amplify |
| • Redux, Redux Toolkit (RTK) | • Node JS, Express | • CloudFront, CloudWatch |
| • Webpack, Vite, Babel | • Strapi | • SQS, EventBridge |
| • Styled Components, Tailwind CSS | • MySQL, PostgreSQL | • Python |
| • Figma, HTML, CSS | • Git (Bitbucket, GitHub) | • Nginx, Redis |

Awards

- Best Newbie (Fresher) Award in the Second Month @ [DemandHelm](#).
- Glow Rock Award of the Month for Best Performance @ [DemandHelm](#).
- Tenet Superhero of the Month @ [DemandHelm](#).

Projects

- **Managing Azure VM Instance with Telegram Bot (Python)**
 - Created REST APIs to turn On/Off Azure VM Instance using Azure Logic Apps and System managed Identities.
 - Then built a Telegram Bot and configured its commands on Python using external library (pyTelegramBotApi) to call those API according to the command given by the user who is authorized to perform the action.
- **Custom Wordle (Next JS + Tailwind CSS)**
 - Developed a Custom Wordle game where the user can play wordle with 4 categories (Fruits, Country, Animal, Bollywood)
 - Along with letter hints, there are also hints about the attributes of the chosen category. Extra hint after 5 wrong attempts.